# Computational Physics Homework 3

**Note:** You can reuse code examples shown in the lecture.

**Q1 [25 pts]:** Exercise 5.9 on the text book (click and see here:chapt5, page 172).

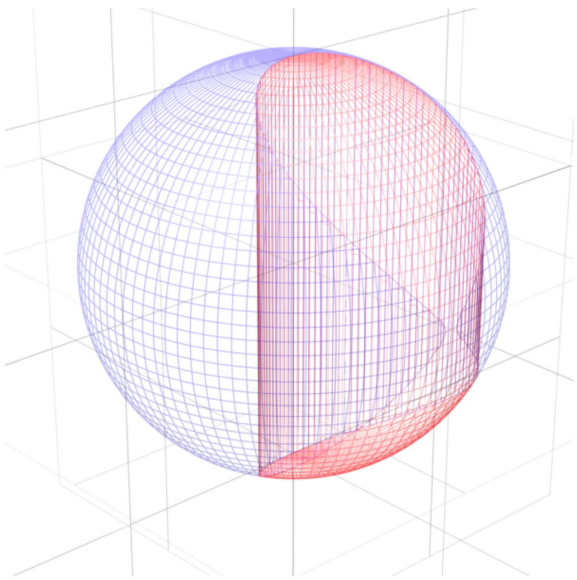
**Q2 [25 pts]:** Exercise 5.12 on the text book (click and see here:chapt5, page 181).

**Q3 [25 pts]:** Calculate the following integral numerically:

$$I = \int_0^1 \frac{x^{-1/4}}{e^x + 2} dx \tag{1}$$

1, Please use the following 2 methods to do the integration with the accuracy of $10^{-5}$: Adaptive Simpson's rule and Gaussian Quadrature.

2, Use importance sampling method to do the integration and compare the results with the uniform sampling mean value method. Make a plot for the result as a function of sampling point number. You need to show the deviation for choosing the weight function and transforming the integrand.

3, Use MCMC (Markov chain Monte Carlo) to calculate the results and show the final result with error.

**Q4 [25 pts]:** Estimate the volume of the intersection of a sphere and a cylinder, using Monte Carlo approach. The sphere has radius 1 and is centered at the origin. The cylinder has radius 1/2, and its axis is perpendicular to the x axis and goes through the point (1/2, 0, 0). (See the following figure ) Report your uncertainty.